Problem R28 (Revised): Viscous Flow in Combined Pipe and Hull System

Scenario

A research vessel uses a hybrid propulsion system composed of:

- a conventional propeller system, and
- a water discharge system that ejects seawater through two **serial** pipe sections (see sketch).

The system is designed such that all pipe flow is **tripped to turbulent regime at the inlet**. The outer hull flow is also assumed **fully turbulent**, avoiding laminar-transition complexity.

Assume seawater at density $\rho = 1025\,\mathrm{kg/m^3}$ and kinematic viscosity $\nu = 1.1 \times 10^{-6}\,\mathrm{m^2/s}$.

Pipe Sections:

- Section A: length $L_A = 30\,\mathrm{m}$, diameter $D_A = 0.3\,\mathrm{m}$, roughness $k_A = 0.3\,\mathrm{mm}$
- Section B: length $L_B = 20\,\mathrm{m}$, diameter $D_B = 0.2\,\mathrm{m}$, roughness $k_B = 1.2\,\mathrm{mm}$

Total volumetric flow rate: $Q = 0.08\,\mathrm{m^3/s}$

Hull Geometry:

- Length $L = 85\,\mathrm{m}$
- Draft $T = 6\,\mathrm{m}$
- Roughness $k_s = 4\,\mathrm{mm}$

Assume the **wetted area** is approximately $A = 2 \cdot L \cdot T$ (i.e. both vertical sides only).

Ship velocity: $V = 9\,\mathrm{m/s}$

Propeller efficiency: $\eta_S = 0.78$

Tasks

a) Calculate the **mean velocity** in each pipe section based on Q and cross-sectional area.

b) Compute the **Reynolds number** for each pipe and confirm turbulent regime.

c) Using the Darcy-Weisbach equation, determine the **total pressure loss** Δp_V over the pipe system:

$$\Delta p_V = \sum_i \left(\lambda_i \cdot \frac{L_i}{D_i} \cdot \frac{\rho}{2} \cdot \bar{u}_i^2 \right)$$

where the friction factor λ for **fully rough turbulent flow** is given by:

$$\frac{1}{\sqrt{\lambda}} = 1.74 - 2\log_{10}\left(\frac{2k}{D}\right)$$

Hint: Alternatively, you may use the pipe friction diagram provided (Lecture Notes p.119).

d) Compute the **friction coefficient** for the hull using:

$$c_f = \left(1.89 + 1.62 \cdot \log_{10}\left(\frac{L}{k_s}\right)\right)^{-2.5}$$

and determine the **total frictional resistance** over the wetted area.

Assume outer flow is fully turbulent. You may also refer to the flat plate diagram (Lecture Notes p.164).

e) Calculate the **total effective power** required to:

- overcome the pipe pressure losses, and
- overcome hull friction resistance.

Then determine the **required engine power** accounting for the propeller efficiency:

$$P = \frac{F \cdot V + Q \cdot \Delta p_V}{\eta_S}$$

If unable to solve previous tasks, assume: $\lambda_A = 0.025$, $\lambda_B = 0.035$, $c_f = 0.0035$

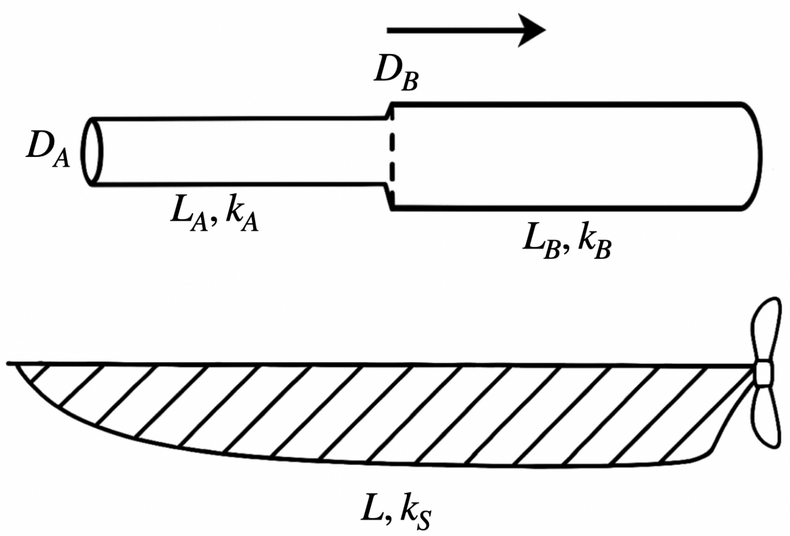

Figure 1: Sketch of the vessel and serial pipe discharge system (not to scale)